

MIRABELLE FENG

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EDUCATION

Carnegie Mellon University

B.S. Computer Science and Arts (BCSA) – GPA 4.0/4.0

Pittsburgh, PA

Aug 2025 – May 2029

- **Coursework:** 15-362 Computer Graphics, 15-210 Parallel & Sequential Data Structures and Algorithms, 15-251 Great Theoretical Ideas in CS, 15-150 Functional Programming, 15-122 Imperative Computation, 21-241 Matrices & Linear Transformations, 60-120 Introduction to Digital Media, 60-212 Creative Coding / Interactivity and Computation

EXPERIENCE

Teaching Assistant, 15-122 Principles of Imperative Computation

Carnegie Mellon University (Spring, Summer, Fall 2026)

Jan 2026 – Present

Pittsburgh, PA

- Lead two weekly recitations and hold office hours for a course serving 1000+ students annually, teaching C programming, data structures, algorithms, and systems concepts
- Since Summer 2026, serve as **grading group lead**, coordinating TAs during staff grading sessions; study rubrics in advance, resolve grading questions, and calibrate scoring to the concepts assessed
- Grade exams and assignments, debug student C code during office hours, and playtest quizzes and homeworks to identify correctness and clarity issues before release

Software Developer, ScottyLabs

Carnegie Mellon University

Aug 2025 – May 2026

Pittsburgh, PA

- Built modular, reusable React and Tailwind CSS components for a campus social app prototype, supporting rapid feature iteration across the team
- Implemented responsive UI from Figma files in collaboration with the design team

PROJECTS

Scotty3D Graphics Engine | C++, 15-362 Computer Graphics

Aug 2026 – Present

- Built a full software rasterization pipeline in C++: scene-graph transforms, diamond-exit line drawing, and bounding-box triangle coverage with top-left tie-breaking, plus depth testing and alpha blending
- Implemented perspective-correct barycentric interpolation with UV derivatives, trilinear mip-mapping (generation, sampling, and LOD selection), and configurable supersampling with a custom sample pattern
- Implementing functionality in geometry meshes, including adding, deleting, and transforming vertices, edges and faces

file fisher | Unity, C#, pixel art | blueherr1ng.itch.io/file-fisher

Aug 2026

- Published a 2D pixel-art desktop idler in which a cat pulls random files out of the user's Downloads folder on a user-set interval, as passive file management for people who find organizing files daunting
- Built the basket UI (open, remove, or send a caught file to the trash) and a settings panel controlling fishing frequency; shipped a signed macOS build to itch.io
- Solo project: wrote all C# game logic and drew all pixel art assets

fish.bowl | TouchDesigner, MediaPipe, real-time 3D

Apr 2026

- Built a hand-tracked interactive installation: a clenched fist spawns a fish into a 3D bowl, and an extended index finger drives a directional "wind" force that pushes fish through the water and can launch them out of the bowl
- Authored the full node network, including 3D scene, models, forces, and gravity restore behavior, on top of the MediaPipe hand-landmark library

TECHNICAL SKILLS

Languages: C, C++, C#, Python, SML, JavaScript, SQL

Graphics & Creative: p5.js, WebGL, Unity, Blender, TouchDesigner, MediaPipe, Photoshop, Illustrator, InDesign, Premiere

Tools: Git/GitHub, VS Code, Figma, Linux/Unix Shell

Activities: Game Creation Society (programmer, *Fruit and Flies*); Computer Graphics Club